

Data Minors

Calculating win probabilities and rankings of women's basketball teams

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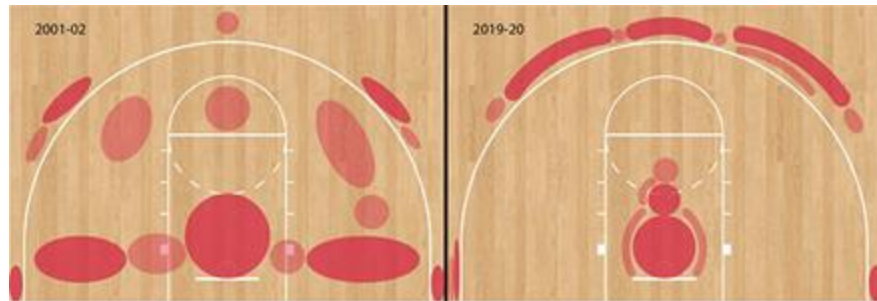


Introduction:

The Wharton High School Data Science Competition introduced us to a unique problem: accurately evaluating a given women's basketball team's strength from data relating to over 5200 matches taken in 2021-2022.

The task included three parts:

1. Rank the top 16 teams in their corresponding regions.
 1. 139 teams across 3 regions, including North, South and West.
1. Predict the winning probability of 10 matchups for the eastern region.
 1. 20 teams where all games were held at a neutral site, thus no home-field advantage.
1. The final stage was to summarise and explain our methodology, which involved:
 1. The tools and techniques we used in terms of software and statistical methods.
 2. Prediction methods to give us the rankings and win percentages.



Change in shooting habits, realisation that 3 pointers are very efficient sources of points

Image source: (Obbie, Kellie. "NBA Teams Use Data Science to Engineer the Perfect Team" 3dfindit)

Background:

Analytics in basketball started off as basic data and minimal datasets in the early 1900s, such as rebounds, points, and assists with minimal fields. This has now progressed, and is now being used by all 30 teams in the NBA. The Phoenix Suns for example, use sensors in addition to data analysis in their 53,000-square-foot practice centre. Equipped with 150 cameras, nodes and sensors, the centre can see subtle changes in movement, and also watch ball movement during plays to learn and thus develop their players. (Obbie, Kellie. "NBA Teams Use Data Science to Engineer the Perfect Team" 3dfindit).

However, for predicting outcomes, multiple studies in 2022 have indicated that logistic regression models outperform other models in terms of success rate (70% - 84%) for predicting games when used in their study. (ÇENE, Erhan "Predicting Euroleague Basketball Match Outcomes With Machine Learning Techniques and Revealing the Most Important Game Related Variables" dergipark)

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Approach:

For our data analysis we decided on using VS Code as our preferred IDE. The entirety of the data analysis was done using Python, which, in conjunction with VS Code, allowed us to easily import open-source modules such as *Matplotlib*, *Scikit-learn*, *NumPy*, *SciPy*, *Pandas* & *SHAP*, which were essential to this project.

As a starting point we decided to start off by filtering the data set and collecting the key features, because this would provide us with a strong foundation and allow us to continue processing the data further. We can select these features by plotting the average of that feature against the winrate, and we chose to do this as it gives us a rough estimate of the statistical significance for each feature. Then we use the remaining features to create a logistic regression model because we predicted that this will give us the best combination of features for the best performance.

Models:

1. Elo Model:

1. We reasoned that an Elo model would be helpful to gauge the relative team strength based on their performance in the data. We would then go on to implement the standard elo rating system, with each team starting at 1500

1. Logistic Regression Model:

1. The machine learning model that we settled on was logistic regression, since matchups are a binary classification problem
2. And the logistic regression model would be helpful to see how stats contribute to team performance

1. Combined Model:

1. In order to create a more accurate model, we decided to incorporate the Elo model predictions into our logistic regression predictions using a weighted system. We did this to create a more comprehensive model that accounts not just for team stats but also historical performance and relative skill.

Methods:

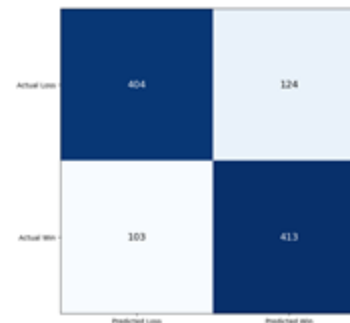
Data set preparation

Ignoring teams with a low game count:

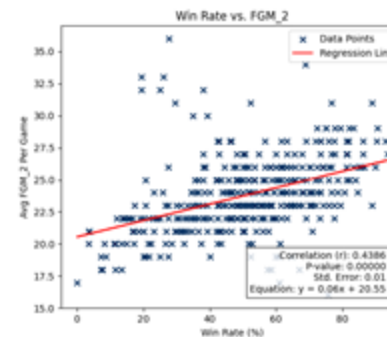
1. Teams that have only played a few games may not provide sufficient data in order to provide a reliable statistical pattern, ultimately skewing the values calculated.
2. We accounted for this by establishing a threshold of 10 games and ignoring any teams that did not meet this requirement.

Feature selection:

1. We initially screened for features that included mostly missing data or gave no valuable information about team performance. These included time zone difference from home stadium with a maximum of 3 hours; attendance; and not a D1 team.
2. For the remaining features, we exported the Google Sheets as a CSV file and, using Python Matplotlib, plotted the team average of that feature against the team win rate (Fig 1.2) to obtain regression and correlation coefficients as well as p-values. Then we narrowed down the list of features by filtering out those with p-values more than 0.05 and selecting the features with the highest regression and correlation coefficients.
3. After this we used the remaining statistics to create a logistic regression model, and through iterative elimination using cross-validation, confusion matrices (Fig 1.1), and classification reports, we found the best combination of features that gave the best performance without overfitting.



▲ Figure 1.1



▲ Figure 1.2

Methods:

Statistical tools & prediction models

Logistic regression model:

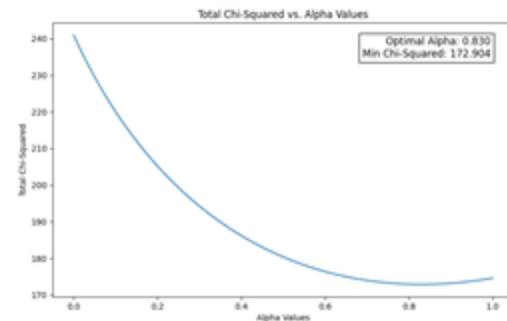
- We tested many different hyperparameters and found the best one for accuracy to be liblinear with L1 regularisation.
- We initially used an 80/20 train-test split to assess model performance; however, to maximise performance against new data, the final model was trained on all the data.
- The final model incorporated 26 features and 13 key performance metrics from each team in the matchup, including (average) *FGA_2*, *FGM_2*, *FGM_3*, *FTA*, *FTM*, *AST*, *BLK*, *STL*, *TOV*, *TOV_team*, *DREB*, *OREB* & *F_personal*.
- To assess model performance, we used 10-fold cross-validation and classification reports. The final model achieved an average precision of 0.77 with an overall score of 0.7674 and a cross-validation score of 0.7603.

Elo Model:

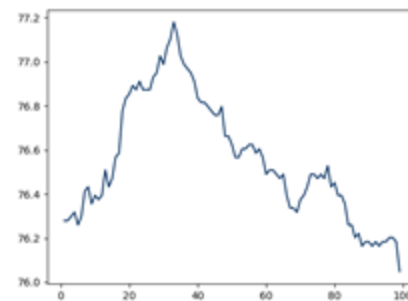
- We used the traditional Elo model, where we plotted the effectiveness of the model against k values (Fig 2.1) to obtain the optimal k value where $R'_a = R_a + k(S_a - E_a)$.

Combined Model:

For each matchup, we calculated the combined win probability using $P(\text{win}) = \alpha \times P_{\text{Elo}} + (1-\alpha) \times P_{\text{Logistic}}$, where α is the weighting coefficient. To find the optimal α , we calculated the Chi-Squared value across α values of 0 to 1 and found an α value of 0.83. This result initially surprised us, but since it is statistically sound, we decided to continue using it. It shows that while the logistic regression model did capture the nuances of team skill, the Elo model gave a stronger indicator of relative team strength.



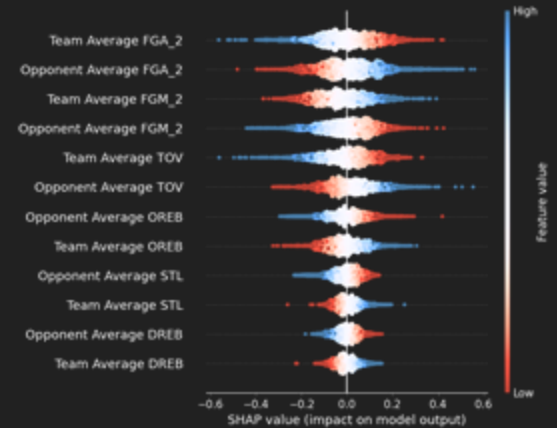
▲ Figure 2.1



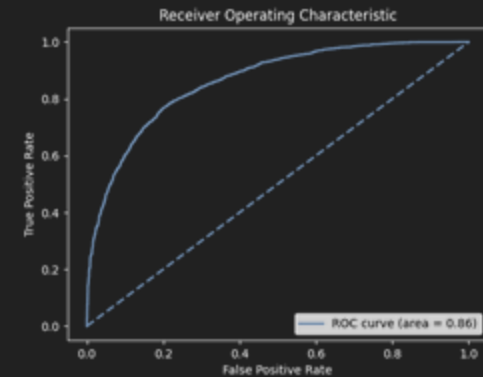
▲ Figure 2.2

Results:

1. Using the same metric as the Logistic regression model, we calculated a cross-validation score of 0.7818, an overall score of 0.7825, and an average precision of 0.78.
 1. Though improvements are marginal, they help take into account a more complete picture of performance.
1. We used *SHapley Additive exPlanations* to see how much each feature contributed to our original logistic model (since our combined model is just a weighted average, this should still be valid).
 1. Our findings shown in Figure 3.1 suggest that team average FGA_2 had the most significant impact on our model; interestingly, the higher the team average FGA_2, the more it negatively impacted the predicted win probability.
 1. This could reflect several basketball realities: teams that rely heavily on 2-point attempts might be less efficient offensively.
1. For a simple view of our results, we plotted receiver operator characteristic curves (Fig 3.2) for true and false positive rates; the area under the curve was calculated to be 0.86, showing it is performing well in its ability to classify data correctly, having a high true positive rate and a low false positive rate.



▲ Figure 3.1



▲ Figure 3.2

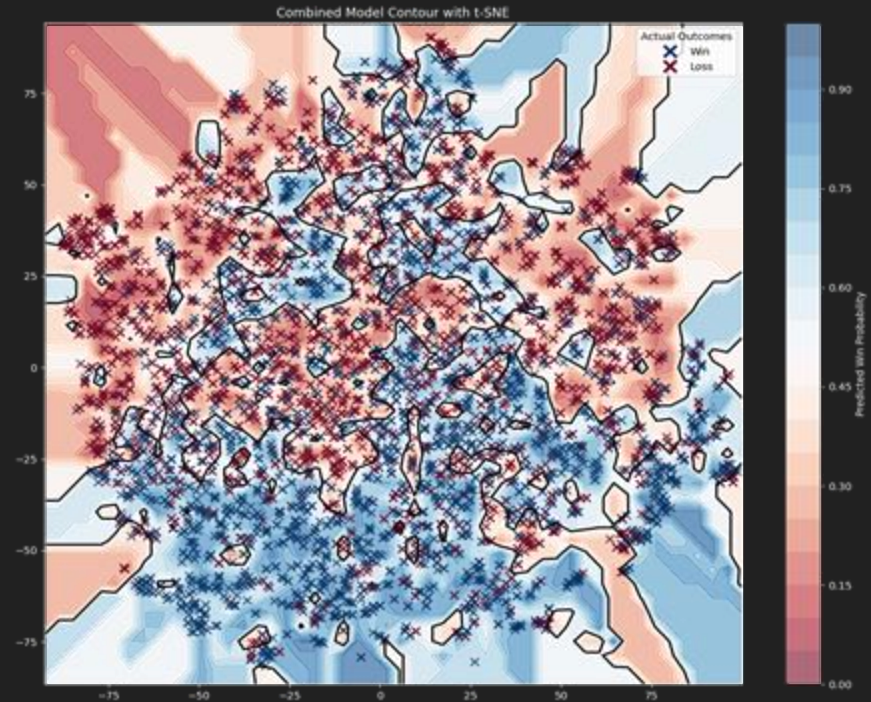
Results:

Due to the high dimensionality of the model and problem, traditional methods of visualisation like two- or three-dimensional decision boundaries yielded seemingly nonsensical graphs. We therefore used dimensionality reduction techniques—principal component analysis (PCA) and t-distributed stochastic neighbour embedding (t-SNE)—to visualise data points and the model's behaviour.

PCA's linearly transformed plots gave little insight, as the data points were too intertwined to draw any meaningful conclusions, which isn't surprising due to basketball's inherent complexity and probabilistic nature.

Although our prediction model uses logistic regression, which is a linear classifier, we chose to use t-SNE. The t-SNE plot (Fig 4) shows that our model is able to group wins and losses moderately accurately, showing that it captures the underlying patterns and non-linear relationships within the data. Though, due to the nature of basketball, the data will not be perfectly separable.

In our t-SNE plot, each data point represents a game outcome from the data. The contour is the theoretical prediction from our model. We generated this by creating a grid of points and, because our model also relies on Elo, assigning a probability to each point based on the model's prediction for the closest real game in the dataset.



▲ Figure 4

Conclusion:

Initially for our rankings, we decided to use a performance-based scoring system; nevertheless, we opted for the combined model, which gave us better results.

Through using this combined model approach, it allowed us to leverage the strengths of both our models, looking not just at relative team skill but also seasonal performance.

We evaluated our models' performance using many different metrics, like receiver operator characteristics curves, confusion matrices and cross-validation.

We used dimensionality reduction techniques such as principal component analysis and t-distributed stochastic neighbour embedding to visualise high-dimensional data and the model's behaviour.

Regardless, there are simply too many features to show them properly in 2 or even 3 dimensions, and while dimensionality reduction techniques do allow us to somewhat visualise the data, they do not show the full picture.

